Case Study 2: Technical Presentation

STAT 946 – Case Studies in Data Science

Your Name Teammate Name Teammate Name

2026-03-09

Section 1

Presentation Goals

Objective of This Presentation

We aim to demonstrate that:

- The analysis is scientifically justified

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- The modeling pipeline is technically reproducible

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- The analysis is scientifically justified
- The modeling pipeline is technically reproducible
- Assumptions and uncertainty are explicitly understood
- Computational decisions are justified
- The approach meaningfully extends Case Study 1

Section 2

Problem Context

Scientific Objective

Goal:

Predict future air pollution concentration

$$y_{t+h,s}$$

using historical pollution and meteorological drivers.

Where:

- t: time index

Output:

$$p(y_{t+h,s} \mid \mathcal{D}_{\leq t})$$

predictive distribution rather than point estimate.

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Where:

- t: time index
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- s: monitoring location

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predictive distribution rather than point estimate.

Data Structure

Dataset characteristics:

- Temporal resolution: hourly observations

Challenges:

Data Structure

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- Multiple monitoring stations

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- Strong seasonality

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- Nonlinearity
- Distributional skewness

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Dataset characteristics:

- Temporal resolution: hourly observations
- Multiple monitoring stations
- Strong seasonality
- Missing observations
- Extreme pollution events

Challenges:

- Temporal dependence
- Nonlinearity
- Distributional skewness
- Covariate interaction effects

Section 3

Case Study 1 Baseline

Previous Approach

Baseline model:

- Describe CS1 model briefly here

Strengths: - Fast - Interpretable - Stable training

Limitations:

Previous Approach

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Limitations:

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Strengths: - Fast - Interpretable - Stable training

Limitations:

- Limited nonlinear modeling
- Weak long-term dependency capture
- Poor extreme-event prediction
- Limited uncertainty characterization

Section 4

Motivation for Case Study 2

Why a More Advanced Method?

Key limitation addressed:

Traditional models assume simplified temporal relationships.

Air quality dynamics exhibit:

- long memory

Therefore we introduce a computationally intensive model capable of learning complex temporal structure.

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Why a More Advanced Method?

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Traditional models assume simplified temporal relationships.

Air quality dynamics exhibit:

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- regime changes

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Section 5

Methodology

Model Overview

We model forecasts using

$$\hat{y}_{t+h,s} = f_\theta(x_{\leq t,s}, y_{\leq t,s})$$

where:

- f_θ is a deep sequence model

(Replace with LSTM / TCN / TFT description)

Model Overview

We model forecasts using

$$\hat{y}_{t+h,s} = f_\theta(x_{\leq t,s}, y_{\leq t,s})$$

where:

- f_θ is a deep sequence model
- parameters learned via gradient optimization

(Replace with LSTM / TCN / TFT description)

Statistical Assumptions

Assumptions include:

- Conditional dependence on past observations

Limitations arise if these assumptions fail.

Statistical Assumptions

Assumptions include:

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- Stationarity within training window

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- Covariates available at forecast time

Limitations arise if these assumptions fail.

Statistical Assumptions

Assumptions include:

- Conditional dependence on past observations
- Stationarity within training window
- Covariates available at forecast time
- Measurement noise treated as stochastic error

Limitations arise if these assumptions fail.

Uncertainty Modeling

Predictive uncertainty estimated via:

- quantile regression / probabilistic output / ensemble

Example:

$$P(y_{t+h} \leq q_\tau) = \tau$$

Produces calibrated prediction intervals.

Section 6

Computational Implementation

Training Procedure

Optimization:

- Loss function: specify here

Training setup:

Runtime considerations:

Training Procedure

Optimization:

- Loss function: specify here
- Optimizer: Adam

Training setup:

Runtime considerations:

Training Procedure

Optimization:

- Loss function: specify here
- Optimizer: Adam
- Learning rate scheduling

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Optimization:

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Training setup:

- GPU computation (Google Colab)

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- Memory constraints

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- Loss function: specify here
- Optimizer: Adam
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- Early stopping

Training setup:

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- Batch training
- Multiple epochs

Runtime considerations:

- Training time
- Memory constraints
- Model scalability

Section 7

Validation Strategy

Leakage-Free Design

Time-respecting split:

- Training period

No future information used during training.

Forecast evaluated strictly forward in time.

Leakage-Free Design

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- Validation period

No future information used during training.

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Leakage-Free Design

Time-respecting split:

- Training period
- Validation period
- Test period

No future information used during training.

Forecast evaluated strictly forward in time.

Hyperparameter Tuning

Tuned parameters:

- learning rate

Search strategy:

Validation set only used for tuning.

Hyperparameter Tuning

Tuned parameters:

- learning rate
- sequence length

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Hyperparameter Tuning

Tuned parameters:

- learning rate
- sequence length
- hidden dimension

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Hyperparameter Tuning

Tuned parameters:

- learning rate
- sequence length
- hidden dimension
- dropout rate

Search strategy:

Validation set only used for tuning.

Hyperparameter Tuning

Tuned parameters:

- learning rate
- sequence length
- hidden dimension
- dropout rate

Search strategy:

- random or grid search

Validation set only used for tuning.

Section 8

Evaluation Metrics

Point Forecast Accuracy

Metrics:

- RMSE

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

Point Forecast Accuracy

Metrics:

- RMSE
- MAE

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

Uncertainty Evaluation

We evaluate:

- Prediction interval coverage

Goal:

High coverage with minimal interval width.

Uncertainty Evaluation

We evaluate:

- Prediction interval coverage
- Interval sharpness

Goal:

High coverage with minimal interval width.

Uncertainty Evaluation

We evaluate:

- Prediction interval coverage
- Interval sharpness
- Calibration reliability

Goal:

High coverage with minimal interval width.

Section 9

Results

Performance Comparison

Compare against Case Study 1:

- accuracy improvement

Insert results table or figure.

Performance Comparison

Compare against Case Study 1:

- accuracy improvement
- robustness during extreme events

Insert results table or figure.

Performance Comparison

Compare against Case Study 1:

- accuracy improvement
- robustness during extreme events
- horizon stability

Insert results table or figure.

Diagnostic Analysis

Diagnostics performed:

- residual analysis

Observed failure modes:

Diagnostic Analysis

Diagnostics performed:

- residual analysis
- temporal error trends

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Observed failure modes:

- peak underestimation

Diagnostic Analysis

Diagnostics performed:

- residual analysis
- temporal error trends
- seasonal performance comparison

Observed failure modes:

- peak underestimation
- regime transition difficulty

Section 10

Technical Value Added

Improvements Over Case Study 1

New capabilities:

- nonlinear temporal learning

Trade-offs introduced:

Improvements Over Case Study 1

New capabilities:

- nonlinear temporal learning
- probabilistic forecasting

Trade-offs introduced:

Improvements Over Case Study 1

New capabilities:

- nonlinear temporal learning
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- improved extreme-event prediction

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- higher runtime

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- nonlinear temporal learning
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- reduced interpretability

Improvements Over Case Study 1

New capabilities:

- nonlinear temporal learning
- probabilistic forecasting
- improved extreme-event prediction

Trade-offs introduced:

- higher runtime
- reduced interpretability
- increased compute requirements

Section 11

Limitations

Statistical Risks

Potential risks:

- overfitting

These define future research directions.

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- overfitting
- covariate shift

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Potential risks:

- overfitting
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- sensor bias

These define future research directions.

Statistical Risks

Potential risks:

- overfitting
- covariate shift
- sensor bias
- model instability under unseen regimes

These define future research directions.

Section 12

Reproducibility

Repository Structure

Reproduction steps:

1. Install dependencies

Artifacts stored:

Repository Structure

Reproduction steps:

1. Install dependencies
2. Load processed dataset

Artifacts stored:

Repository Structure

Reproduction steps:

1. Install dependencies
2. Load processed dataset
3. Run training notebook

Artifacts stored:

Repository Structure

Reproduction steps:

1. Install dependencies
2. Load processed dataset
3. Run training notebook
4. Execute evaluation pipeline

Artifacts stored:

Repository Structure

Reproduction steps:

1. Install dependencies
2. Load processed dataset
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Artifacts stored:

- trained weights

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Reproduction steps:

1. Install dependencies
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Artifacts stored:

- trained weights
- prediction outputs

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Reproduction steps:

1. Install dependencies
2. Load processed dataset
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Artifacts stored:

- trained weights
- prediction outputs
- evaluation metrics

Computational Reproduction

Execution platform:

- Google Colab GPU

Approximate runtime:

Random seeds fixed where possible.

Computational Reproduction

Execution platform:

- Google Colab GPU

Approximate runtime:

- preprocessing

Random seeds fixed where possible.

Computational Reproduction

Execution platform:

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Approximate runtime:

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- training

Random seeds fixed where possible.

Computational Reproduction

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Approximate runtime:

- preprocessing
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- evaluation

Random seeds fixed where possible.

Section 13

Key Takeaways

Summary

- Advanced model improves forecasting performance

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- Uncertainty estimates are quantitatively validated

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- Method addresses limitations from Case Study 1

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- Advanced model improves forecasting performance
- Uncertainty estimates are quantitatively validated
- Method addresses limitations from Case Study 1
- Computational cost justified by predictive gain

Section 14

Questions

Q&A

We welcome discussion on:

- assumptions

Q&A

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- assumptions
- validation design

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- uncertainty interpretation

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- scalability

Section 15

Appendix

Additional Model Details

Include:

- architecture diagram

Additional Model Details

Include:

- architecture diagram
- hyperparameters

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Include:

- architecture diagram
- hyperparameters
- ablation studies

Additional Diagnostics

- calibration curves

Additional Diagnostics

- calibration curves
- residual plots

Additional Diagnostics

- calibration curves
- residual plots
- site-level performance

Reproducibility Notes

Large artifacts stored externally if required.

Instructions documented in repository README.